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MODELS

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Aggregating Average Treatment Effects on the Treated in Difference-in-Differences Models
Partha Deb, Edward C. Norton, Jeffrey M. Wooldridge, and Jeffrey E. Zabel
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ABSTRACT

For difference-in-differences methods, there has been great attention to obtaining consistent estimates of treatment effects, especially when the treatment effects are heterogeneous. However, there has been little discussion of the importance of weights in aggregating those treatment effects into an overall average treatment effect on the treated. There are many possible ways to aggregate estimated cohort-time treatment effects. We show that the standard software used to estimate Callaway and Sant’Anna’s method uses weights that are not just the number of treated observations in treated years. Instead, the software uses weights that include the number of observations in the reference pre-period instead of only the number of observations in the treated periods. We discuss why the aggregation weights matter and under what circumstances the weights make the most difference.

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1 Introduction

Many recent papers on difference-in-differences methods explain how to estimate consistent treatment effects when treatment start times are staggered and those treatment effects are heterogeneous over cohorts and time (e.g., [de Chaisemartin and D’Haultfoeuille, 2020](#); [Callaway and Sant’Anna, 2021](#); [Goodman-Bacon, 2021](#); [Sun and Abraham, 2021](#); [Wooldridge, 2025](#); [Borusyak et al., 2024](#); [Deb et al., 2025](#)). These methods typically estimate numerous treatment effects, over cohorts (or groups) and time. These many treatment effects can be difficult to report and interpret. Therefore, researchers often report summary measures, such as the overall average treatment effect on the treated (ATET). Or they report the treatment effects averaged over time since start of treatment, or by cohort, or by calendar time, or for an event study (including during the pre-treatment periods). These average treatment effects are a weighted average of the estimated treatment parameters.

We show that the popular software that estimates the [Callaway and Sant’Anna \(2021\)](#) difference-in-differences method uses weights based on the number of observations in both pre-treated periods and treated periods, for repeated cross-sectional data when aggregating the estimated treatment effects to the ATET. The Stata commands `csdid` and `hdidregress` use these weights for repeated cross-sectional data. Although there are many ways to estimate an aggregated ATET, the one most commonly described in the literature only involves the number of treated observations corresponding to that cohort and time. Instead, the software that implements the Callaway and Sant’Anna method also include the number of observations from the reference pre-period in the calculation of each of the weights. Although this issue does not matter for estimating ATETs if the data set is a perfectly balanced panel, it does matter when there is a repeated cross section with an unequal number of observations in each period, as is often the case in empirical papers.

This paper has four goals. First, to describe cases where different methods yield identical parameter estimates of treatment effects over cohorts and time. Second, to explain when those same methods can yield non-identical parameter estimates (e.g., due to differences in the model specification of covariates). Third, to show that even when different methods yield identical parameter estimates, they can provide different estimates of the overall ATET

through the use of different weights applied to the underlying estimates of the individual dynamic treatment effects. In our empirical example with simulated repeated cross-sectional data, we compare the different aggregated treatment effects estimated automatically by the software implementing the Callaway and Sant’Anna method compared to other methods. Fourth, to explain when the weighting matters the most and when it does not make much of a difference. Although the differences could be large, it depends on the study design and data. There is no difference when the data are a balanced panel. There is often a small difference for repeated cross sections. The difference is largest when the sample sizes decrease during the treated periods.

To show this, we create a simulated data set where we know the true ATETs and, consequently, the aggregate ATET for any set of weights. Then we estimate the heterogeneous treatment effects and ATET using both the [Callaway and Sant’Anna \(2021\)](#) and our own FLEX (**F**lexible **L**inear **E**stimator with covariates **X**) method ([Deb et al., 2025](#)). For Callaway and Sant’Anna, we use the Stata software commands to calculate the ATET. We show that, although both methods produce identical estimates of ATET at the cohort-time level, the FLEX method and the Callaway and Sant’Anna method estimate different ATETs, as implemented by the Stata statistical software. We conclude that the software documentation should be clarified so that users know what ATET is calculated. This particular issue of aggregation weights has not been raised in recent summary papers about difference-in-differences methods ([Baker et al., 2025](#); [de Chaisemartin and D’Haultfœuille, 2023](#); [Freedman et al., 2023](#); [Roth et al., 2023](#)).

2 Same Estimated Coefficients

[Wooldridge \(2025\)](#) provides a unified approach to regression-based difference-in-differences estimators when there is staggered entry and heterogeneous treatment effects (both across cohorts and over time). In brief, he allows a full set of treatment effects on the treated. The treatment effects can be by cohort, for each time period, and can be interacted with covariates. His general approach accommodates either lags-only or lags-and-leads models, depending on whether the treatment effects are only estimated in the periods with treatment or in all periods. Wooldridge assumes no conditional anticipation, conditional parallel trends,

and linearity.

Wooldridge (2025) discusses how many different estimators yield numerically identical estimated treatment effects. Here are a few illustrative examples. The imputation method (Borusyak et al., 2024) is identical to pooled OLS with lags only for either panel data or for repeated cross-sectional data (Deb et al., 2025). For models with lags and leads, a sufficiently flexible OLS regression can reproduce the same results as the Sun and Abraham (2021) event study or the Callaway and Sant’Anna (2021) method with the regression adjustment. We focus primarily on FLEX and Callaway and Sant’Anna to illustrate our main points. Furthermore, we estimate models without covariates and with heterogeneous treatment effects at the cohort-time level.

3 Different Estimated Coefficients

Despite the many ways that seemingly different methods can yield numerically identical estimated treatment effects, most often the methods yield numerically different results even when applied to the same data set. First, some methods estimate treatment effects at different levels. For example, if multiple states enact a policy at the same calendar time, one could model treatment effects at either the state (group) level or at the cohort level, where cohorts are defined as all groups starting in the same period. The former is more flexible and allows for more heterogeneity, whereas the latter is more efficient if the group-level treatment effects are homogeneous within cohorts. In this paper, for simplicity and compatibility across different software packages, we estimate treatment effects at the cohort-time level.

Second, different models may treat covariates differently. Covariates can be entirely omitted, as in our example later in this paper. They can be included additively. Or they can be interacted with treatment effects (both across cohorts and over time), allowing for an extremely flexible model specification (Wooldridge, 2025; Deb et al., 2025). In this paper, we do not include covariates in our example to focus on a simpler model in which any reasons for differences in ATETs cannot be due to covariates.

Third, there can be differences in whether the model estimates lags only or lags and leads (Deb et al., 2025). A *lags-only model specification* only estimates differences between treatment and controls in the post periods but assumes that the only differences between

treatment and control groups in the pre-periods are up to a level fixed effect. A *lags-and-leads model specification* estimates additional effects for each treatment group in each pre-period. If the parallel trends assumption is correct, then, the results of the two models should be similar. If the parallel trends assumption is incorrect, then the lags-only model will incorrectly impose homogeneity in the pre-periods between treatment and control groups (up to an additive intercept). Callaway and Sant’Anna (2021) estimate a lags-and-leads model specification; FLEX can estimate either a lags-only or lags-and-leads model specification. In this paper, we estimate models with lags and leads, again for simplicity and so that the estimated treatment effects are identical.

Fourth, Callaway and Sant’Anna (2021) propose a doubly-robust method that estimates both the outcome and the treatment, and as long as one of the models is correctly specified, then the results are consistent. In our simple example with no covariates, the regression adjustment and the doubly robust versions of Callaway and Sant’Anna are numerically identical, although that is not usually the case when the model includes covariates. In this paper, when we estimate models with Callaway and Sant’Anna’s estimator, we do not use their doubly robust estimator.

4 Aggregation To Get The Overall ATET

There are many ways to aggregate cohort-time treatment effects. In this section we derive the aggregation formulas that are used by most difference-in-differences estimation packages and compare that to what Callaway and Sant’Anna implement in their software packages.

Define cohorts by the period when units in that cohort first become treated. Let D_c be a binary variable that is equal to one if a unit in that cohort is first treated in period c , for $c = 2, \dots, T$. Here we assume that no cohorts start treatment until at least the second period because we need observations in at least one pre-period for each cohort. We allow for the case of multiple pre-treatment periods by setting $D_c = 0$ for periods prior to the first treatment period. There is at least one cohort that is never treated.

We allow for heterogeneous treatment effects by both cohorts and time periods. In the potential outcomes framework, the average treatment effect on the treated (ATET) for individuals in cohort c (first treated in period c) in time period t for $t \geq c$ is defined as

$$ATET(c, t) = E[Y_t(1) - Y_t(0) | D_c = 1] \quad (4.1)$$

where $Y_t(1)$ and $Y_t(0)$ are the values of potential outcome Y in period t when treated and not treated, respectively. The $ATET(c, t)$ are referred to as the cohort-time treatment effects.

Let N be the total number of units sampled (treated and untreated) and N_{ct} be the number of units that are first treated in period c that are sampled in period t . In the panel data case N_{ct} is the same in all periods t for cohort c . In the repeated cross section case, N_{ct} will differ in each period.

The overall ATET, weighted by the number of treated observations in treated periods, across all cohorts and time periods is the weighted average of the $ATET(c, t)$'s, where the weights are the number of observations for cohort c in treated time periods (time $t \geq c$), and C is the set of treated cohorts.

$$ATET = \left(\frac{1}{\sum_{c \in C} \sum_{t \geq c} N_{ct}} \right) \sum_{c \in C} \sum_{t \geq c} N_{ct} ATET(c, t) \quad (4.2)$$

Callaway and Sant'Anna (2021) have a lengthy discussion of ways to aggregate cohort-time treatment effects using different weights. In section 3.2 of their paper, they present several formulas “for aggregating group time average treatment effects into an overall effect of participating in the treatment.” Their first aggregation formula (3.10 in their paper, slightly modified here by changing g to c , G to C , and ATT to $ATET$) uses weights proportional to the number of treated observations for a treated group (or cohort) during a treated time period.

$$\theta_W^O = \frac{1}{\kappa} \sum_{c \in C} \sum_{t=2}^T \mathbf{1}\{t \geq c\} ATET(c, t) P(C = c | C \leq T)$$

where κ is the sum of the weights, $\mathbf{1}\{t \geq c\}$ is an indicator for post periods, $ATET(c, t)$ are the treatment effects for cohort c at time t , and $P(C = c | C \leq T)$ are the weights for each treated cohort. Callaway and Sant'Anna (2021) explain on page 211 that θ_W^O is a weighted average of each ATET that puts more weight on ATETs's with larger group sizes. This is also the formula for the weighted average ATET shown in the Stata documentation of `hdidregress postestimation`.

Callaway and Sant’Anna’s weights $P(C = c|C \leq T)$ are not a function of time even though the treatment effects $ATE(c, t)$ are a function of both cohort and time. The condition that $C \leq T$ simply restricts weights to cohorts that are treated (never treated cohorts have $C = \infty$). So in their formula, Callaway and Sant’Anna have a single weight for each treated cohort in all treated periods.

But what is calculated in Stata’s command `hdidregress postestimation` and in the user-written package `csdid` is instead the following, *with an additional term $N_{c,c-1}$ in both the numerator and the denominator, representing the number of observations for cohort c in the pre-treated period ($t = -1$):*

$$ATE = \left(\frac{1}{\sum_{c \in C} \sum_{t \geq c} (N_{ct} + N_{c,c-1})} \right) \sum_{c \in C} \sum_{t \geq c} (N_{ct} + N_{c,c-1}) ATE(c, t) \quad (4.3)$$

In general, equation 4.3 does not equal equation 4.2. Only in the pure balanced panel data case will equation 4.3 simplify to equation 4.2, because $N_{ct} = N_{c,c-1}$ across all time periods for cohort c . Furthermore, these weights differ by both cohort and by time, unlike their formula.

Callaway and Sant’Anna (2021) observe that different weights will yield different overall averages, except in the unusual case that all ATETs are identical. They specifically mention ATETs, weighted by the number of treated observations in treated periods, and equal-weighted ATETs as having a meaningful interpretation. At no time do they mention including the number of observations in any pre-treatment periods as part of the weights. Nor is that mentioned in the Stata documentation of `hdidregress postestimation` or `csdid`.

Callaway and Sant’Anna also say that one might consider alternative weights, for example, putting equal weight on each cohort so as not to put more weight on cohorts that participate longer.

We agree that there are different ways to estimate aggregated treatment effects using different weights, and that those different aggregated ATETs will have different interpretations and different numerical values (except in the extreme case of homogeneous treatment effects across all cohorts and periods). Therefore, we agree with the theoretical part of the Callaway and Sant’Anna (2021) paper. Our concern is that the implementation of their code yields a weighted average that is hard to interpret and is not what researchers expect.

In addition to the overall ATET, researchers often also calculate other average treatment effects, for example by cohort, by event time, or by calendar time (also discussed by [Callaway and Sant’Anna \(2021\)](#)). The aggregation issues discussed in this section also apply to each of those average treatment effects.

5 Empirical Example With Simulated Data

5.1 Simulated Data

To illustrate our main points, we use simulated data that has features of data sets that are often used for difference-in-differences analysis of repeated cross sections with staggered treatment start times. Our simulated data has 50 groups, representing 50 states. There are nine time periods, representing nine years, with three different treatment cohorts starting in years four, six, and eight, respectively. There are 10 groups in each of the three treatment cohorts, leaving 20 control groups.

The continuous outcome Y varies by group and by time and has an overall mean of 4.6 (see Table 1). There are additive group fixed effects, representing different overall means of Y , conditional on the group. Therefore, in the initial period, the expected value of the outcome y differs across all groups. There is a common time trend in the outcome for all groups and periods, as well as time-specific fixed effects that adds variation around that common time trend. Therefore, the expected value of the outcome y generally increases over time in the control cohorts, with some annual fluctuation. There is an additive error with a normal distribution. The parallel trends assumption and the no-anticipation assumptions hold.

The advantage of simulated data is that we know the true ATET for any definition of ATET. Therefore, we can show whether the DID aggregation methods produces estimates that are close to the true ATET or not. The three treated cohorts, each comprised of 10 groups, have heterogeneous treatment effects, both across cohorts and over time, by construction. The cohort that starts treatment in period 4 (Cohort 4) has an expected first-treatment-period increase in the dependent variable of 1.5, followed by an increase of 1.0 per year (see Table 2 for the expected treatment effects by cohort and time). The next cohort

starts treatment in period 6 (Cohort 6). It has an expected initial increase in the dependent variable of only 1.0, with an annual increase of 0.7. The last cohort starts treatment in period 8 (Cohort 8). It has an expected initial increase of 0.5, and then an annual increase of 0.4 (for one period). Therefore, the treatment effects vary dynamically across cohorts and across time periods. The numerical values of the expected treatment effects are shown in Table 2. To see the visual representation of the treatment effects by cohort and time compared to the controls, see Figure 1. The predicted values of the outcome Y for each cohort, based on a simple regression with separate time effects by cohort, show the differences by cohort and compared to the control groups.

We point out that while there are group fixed effects for all 50 groups, we estimate heterogeneous treatment effects at the cohort-time level, not the group-time level. This makes it easier to compare results across different software packages.

To calculate the expected treatment effects by cohort and time, we need the number of observations corresponding to each estimated treatment effect (by cohort and time, in the treated periods). The sample size per group and period varies — from around 200 to more than 8,000 — reflecting different populations across the states and having an unbalanced repeated cross section. We need the number of observations to create weights when aggregating the treatment effects.

In this example, we have different numbers of observations by cohort over time, as is typical in a repeated cross section (see Table 3). Overall, the number of observations decreases with time since the start of treatment. The number of treated observations in each cohort and time period is shown in bold. The number of observations in the ever-treated cohorts in the period prior to start of treatment are highlighted in italics. The treatment effects are averaged in proportion to the number of treated observations.

The expected overall ATET is the weighted average of treatment effects shown in Table 2 (treatment effects are in bold) weighted by the corresponding number of observations in Table 3. When using the weights based only the number of treated observations, the expected ATET is 2.8040062. When using the weights that include the number of pre-treated observations, which also include the number of observations in the period just prior to the start of treatment, the expected ATET is 3.259436, for a difference of 0.4554298 (a

16.2% difference). Because our simulated data is constructed with additive error terms, the actual estimated ATET should be close to, but not numerically identical to 2.8040062.

While a difference of 0.455 may not seem like a large difference numerically, we respond with three points. First, we could construct the simulated data to have any size effect that we wanted, large or small, positive or negative. We chose an example with three different cohorts, each with a different initial treatment effects and slope over time, similar to what is found in many empirical studies. We also chose a decline in the number of observations in the treated years, because that is a case where the weighting formula matters the most. Second, we will show that the estimated heterogeneous cohort-time treatment effects are numerically identical, so any difference in the ATET is due entirely to the aggregation weights. Third, the theory is clear on what the correct weights are — the weights should only depend on the number of observations in treated periods. The difference that we find is due to implementation of the statistical code, not due to any disagreement about what the right answer is.

We also show this for aggregated treatment effects besides the overall ATET. We calculate ATETs by cohort, by calendar year, and by time since the start of treatment. All of these ATETs can be calculated automatically for Callaway and Sant’Anna with Stata software, and they all have the same weighting formula that depends in part on the number of observations in the period just prior to the start of treatment.

Our example shows the importance of aggregation weights in the simplest possible way. There are no covariates, such as individual demographics or state characteristics. The Stata code used to generate the data and analyze the results is available upon request.

5.2 DID Methods

We estimated two main different empirical difference-in-differences models to estimate treatment effects by cohort and year. The first is the lags-and-leads FLEX. For FLEX we used the `flexdid` command (Deb, 2025), which is available on Statistical Software Components (SSC) at <https://ideas.repec.org/c/boc/bocode/s459517.html>. The postestimation command `estat atet` calculates the overall ATET. The second is the Callaway and Sant’Anna method with the regression adjustment, which returns identical parameter esti-

mates of the cohort-year treatment effects but different aggregated ATETs. The Callaway and Sant’Anna results were obtained with the Stata command `hdidregress` with the `ra` option and the Stata command `estat aggregation` to get the aggregate ATET. We confirmed that we got numerically identical results for Callaway and Sant’Anna with the Stata command `csdid` followed by `estat simple`.

We also confirmed that the Stata command `did_imputation` for the imputation method of [Borusyak et al. \(2024\)](#) also calculates the exact same ATET as FLEX ([Deb et al., 2025](#)). We can show this for both the lags only model specification and the lags and leads model specification, which requires creating a set of dummy variables for each treated cohort and time. Therefore, the Stata software for the imputation method uses the correct weights for aggregation. In addition, we used the method of [de Chaisemartin and D’Haultfoeulle \(2020\)](#) with their `did_multiplegt` command.

This is not a problem that using the [Callaway and Sant’Anna \(2021\)](#) doubly robust option can solve. Callaway and Sant’Anna with the doubly robust option produces numerically identical parameter estimates of the cohort-time treatment effects, and the software calculates the same ATETs as with the regression adjustment.

5.3 Results

The estimated cohort-time treatment effects are identical for FLEX and for Callaway and Sant’Anna for the model specifications with no covariates, lags and leads, and heterogeneous treatment effects estimated at the cohort-time level. The estimated treatment effects are shown in Table 4 by cohort and time. Not surprisingly, the estimated treatment effects in Table 4 are close to the expected treatment effects due to the data generating process, shown in Table 2. However, the ATET are different between Callaway and Sant’Anna and FLEX because they use different weights, as shown in section 4. FLEX uses weights that are equal to the number of observations in each treated cohort-time (see Table 3). The estimated treatment effects are in Table 4.

For example, in cohort 4 at time 4 the weight is **906**; for cohort 6 at time 6 the weight is **520**; for cohort 8 at time 8 the weight is **192**. Therefore, the weighted average ATET for

FLEX is 2.799203:

$$\begin{aligned}
ATET^{FLEX} = & ((\mathbf{906}) \times 1.540452 + (\mathbf{906}) \times 2.452483 + (\mathbf{755}) \times 3.534971 \\
& + (\mathbf{302}) \times 4.581689 + (\mathbf{453}) \times 5.500610 + (\mathbf{302}) \times 6.517526 \\
& + (\mathbf{520}) \times 0.995921 + (\mathbf{208}) \times 1.532444 + (\mathbf{312}) \times 2.276666 \\
& + (\mathbf{208}) \times 3.035713 + (\mathbf{192}) \times 0.662019 + (\mathbf{128}) \times 0.763577) \\
& / (\mathbf{906} + \mathbf{906} + \mathbf{755} + \mathbf{302} + \mathbf{435} + \mathbf{302} + \mathbf{520} + \mathbf{208} + \mathbf{312} + \mathbf{208} + \mathbf{192} + \mathbf{128})
\end{aligned} \tag{5.1}$$

In contrast, the ATET estimated for the Callaway and Sant’Anna software has different weights. Callaway and Sant’Anna uses the sum of the number of observations in each treated cohort-time plus the number of observations in the year prior to the start of treatment. The additional weight is the same within a cohort for every period. Therefore, the pre-period weights are included not just once, but multiple times.

For example, in cohort 4 at time 4 the weight is **906** plus *1,359*; for cohort 6 at time 6 the weight is **520** plus **624**; for cohort 8 at time 8 the weight is **192** plus **128**. Therefore, the average treatment effect on the treated, which depends on the weights, are different for Callaway and Sant’Anna than for FLEX (see Table 5). The later time periods are weighted relatively more by Callaway and Sant’Anna, resulting in a higher estimated ATET. The weighted average ATET for Callaway and Sant’Anna is 3.254948:

$$\begin{aligned}
ATET^{CS} = & ((\mathbf{906} + 1359) \times 1.540452 + (\mathbf{906} + 1359) \times 2.452483 \\
& + (\mathbf{755} + 1359) \times 3.534971 + (\mathbf{302} + 1359) \times 4.581689 \\
& + (\mathbf{453} + 1359) \times 5.500610 + (\mathbf{302} + 1359) \times 6.517526 \\
& + (\mathbf{520} + 624) \times 0.995921 + (\mathbf{208} + 624) \times 1.532444 \\
& + (\mathbf{312} + 624) \times 2.276666 + (\mathbf{208} + 624) \times 3.035713 \\
& + (\mathbf{192} + 128) \times 0.662019 + (\mathbf{128} + 128) \times 0.763577) \\
& / (\mathbf{906} + 1359 + \mathbf{906} + 1359 + \mathbf{755} + 1359 + \mathbf{302} + 1359 \\
& + \mathbf{435} + 1359 + \mathbf{302} + 1359 + \mathbf{520} + 624 + \mathbf{208} + 624 \\
& + \mathbf{312} + 624 + \mathbf{208} + 624 + \mathbf{192} + 128 + \mathbf{128} + 128)
\end{aligned} \tag{5.2}$$

In addition to the overall aggregate ATET, we also show results for the aggregation of estimated treatment effects for an event study (see Table 6. For most time periods the

aggregated estimates are similar but not identical, for the same reason of different weights. However, for time pre-periods -7 and -6 and for post-periods 4 and 5 (relative to the start of treatment) they are again numerically identical. That is because in this empirical example, only Cohort 8 is observed in pre-periods -7 and -6 and only Cohort 4 is observed in post-periods 4 and 5, so the relative weights between cohorts do not matter.

In results that we do not show, we also find that aggregated treatment effects at the cohort level and at the calendar time level are different because of the different weights. Therefore, this issue of weighting affects not just the overall aggregated ATET, but all aggregated results.

5.4 Simulation

Although we found a meaningful difference between the aggregated overall ATET between FLEX and Callaway and Sant’Anna in one example, we also wanted to get a broader sense of differences across other methods using more simulations. Therefore, we expanded our example in two ways. First, we did 20 simulations which varied in the magnitude of the true treatment effects, the number of observations per cohort and time, and the error terms. The overall study design (9 periods, three treated cohorts, heterogeneous treatment effects by cohort and time) remained the same as the earlier example.

Second, we compared four different estimation methods using publicly available Stata software. These estimation methods include 1) basic homogeneous treatment effects, often called two-way fixed effects (TWFE), estimated by `regress` or `didregress`; 2) [Deb et al. \(2025\)](#) FLEX with lags and leads, estimated by `flexdid`; 3) [Borusyak et al. \(2024\)](#) imputation method with lags and leads, estimated by `did_imputation` with additional dummy variables spanning all treated cohort periods; and 4) [Callaway and Sant’Anna \(2021\)](#), estimated by `hdidregress`. As is well known, the homogeneous TWFE only estimates a single overall treatment effect, essentially combining estimation and aggregation in the process of estimating a single parameter. The other three estimators all estimate identical cohort-time treatment effects in these simulations with no covariates. Therefore, any differences in an estimated overall ATET is due to aggregation.

We found surprising results, as shown visually in [Figure 2](#). Each column refers to results

from one of twenty simulated data sets. The vertical axis is the magnitude of the overall ATET, roughly 5.0. The black dots (connected by a line) are the expected ATETs, weighted by the number of treated observations in treated periods; these vary across simulations because of differences in the expected cohort-time treatment effects and variation in the number of observations per cohort and period, creating different weights for each simulation. The two concentric red circles show that FLEX and BJS imputation aggregate the same cohort-time treatment effects to the same weighted ATET, and those are quite close to the expected values, differing because of the random error term. In other words, these two estimators use the same weights to estimate an overall ATET. As expected, the TWFE ATET is quite different.

Surprisingly, the Callaway and Sant’Anna ATETs are also typically quite far from the expected ATETs. Again, any differences in the aggregated ATET is solely due to aggregation weights, not to the estimated cohort-time treatment effects, which are the same as the other three non-TWFE methods. In fact, in four of the simulations (2, 6, 7, and 19), the CS ATET was further from the expected ATET, weighted by the number of treated observations in treated periods, than the TWFE ATET. We have done numerous other simulations and found that pattern persists. If researchers avoid TWFE because it is biased and often far from the expected value, then they should also be cautious about Callaway and Sant’Anna’s weighted ATET.

In addition, we explored using the method of [de Chaisemartin and D’Haultfœuille \(2020\)](#), estimated by `did_multiplegt`. Their method estimates two-by-two comparisons but they include any not-yet-treated observations as part of the controls. Therefore, [de Chaisemartin and D’Haultfœuille](#) and Callaway and Sant’Anna with the `notyet` option yield identical estimated cohort-time treatment effects. However, they do not return identical overall ATET. [de Chaisemartin and D’Haultfœuille \(2020\)](#) use weights based only on the number of treated observations in treated periods, just like [Borusyak et al. \(2024\)](#) and [Deb et al. \(2025\)](#). In summary, for balanced panel data, [de Chaisemartin and D’Haultfœuille](#) is identical to Callaway and Sant’Anna with the `notyet` option. But for repeated cross sectional data, [de Chaisemartin and D’Haultfœuille](#) is different than the other methods because of the difference in control observations, which affects the estimated cohort-time treatment effects.

6 When Weighting Matters Most

The aggregation weights matter. If the data set was a balanced panel, then the difference in the weighting formulas would balance out, because Callaway and Sant’Anna’s weights would exactly double the weights of the ATET, weighted by the number of treated observations in treated periods,. However, in repeated cross-sectional data, the number of observations typically differs from year to year in a cohort, so it matters whether part of the weights include the number of observations from the pre-treated year.

When will the weighting matter more? In brief, the weighting matters more when there is more heterogeneity in the treatment effects, when there is more heterogeneity in the number of observations, and when those heterogeneities are correlated. The weights produce bigger differences between methods when the number of observations in the pre-period is relatively large compared to the number of observations in the treated periods.

In our empirical example the number of observations in the pre-period is larger than in any of the later treated periods. When those pre-period weights are included, they give more relative weight to the later periods. Because the treatment effects increase over time, the Callaway and Sant’Anna weights put more emphasis on the later periods, which in this example means a higher overall ATET. The heterogeneity in the treatment effects by cohort and time is modest. We could have constructed an example in which the difference in estimated ATETs was either much larger or much smaller.

7 Conclusions

We show that the weights used to aggregate estimated treatment effects at the cohort-time level into an overall ATET matter. The weights are different in the Stata commands `hdidregress` and `csdid` than what [Callaway and Sant’Anna \(2021\)](#) discuss in their theoretical paper. For many simple difference-in-differences model specifications, the choice of estimator yields identical point estimates of the group-time treatment effects. This should result in identical estimates of the average treatment effect on the treated and other aggregated treatment effects (such as overall, by calendar time, by cohort, or by time since start of treatment). However, for the [Callaway and Sant’Anna \(2021\)](#) method, the Stata commands

use weights based in part on the number of observations in the period just prior to the start of treatment. This will yield different estimates of the ATET compared to other difference-in-differences estimators whenever the data set has an unbalanced number of observations. The difference will be greater when the number of observations in the period immediately prior to start of treatment is relatively large and there is heterogeneity across cohorts and over time; the difference will be zero with a balanced panel.

Given the enormous influence and popularity of the [Callaway and Sant'Anna \(2021\)](#) method for estimating heterogeneous treatment effects in difference-in-differences models with staggered start time, and the common use of Stata software, we are concerned that many published results have been calculated with a formula that is not what the researchers intended.

References

- Baker, Andrew C. Callaway, B., Scott, C., Goodman-Bacon, A., and Sant’Anna, P. H. C. (2025). Difference-inDifferences Designs: A Practitioner’s Guide. Technical report.
- Borusyak, K., Jaravel, X., and Spiess, J. (2024). Revisiting Event-Study Designs: Robust and Efficient Estimation. *The Review of Economic Studies*, page <https://doi.org/10.1093/restud/rdae007>.
- Callaway, B. and Sant’Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2):200–230.
- de Chaisemartin, C. and D’Haultfœuille, X. (2020). Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review*, 110(9):2964–2996.
- de Chaisemartin, C. and D’Haultfœuille, X. (2023). Two-way fixed effects and difference-in-differences with heterogeneous treatment effects: a survey. *Econometrics Journal*, 26:C1–C30.
- Deb, P. (2025). Flexdid: Stata module for flexible estimation of difference-in-differences regression with staggered implementation.
- Deb, P., Norton, E. C., Wooldridge, J. M., and Zabel, J. E. (2025). A flexible, heterogeneous treatment effects difference-in-differences estimator for repeated cross-sections.
- Freedman, S. M., Hollingsworth, A., Simon, K. I., Wing, C., and Yozwiak, M. (2023). Designing Difference in Difference Studies With Staggered Treatment Adoption: Key Concepts and Practical Guidelines. NBER Working Paper 31842. <https://www.nber.org/papers/w31842>.
- Goodman-Bacon, A. (2021). Difference-in-differences with Variation in Treatment Timing. *Journal of Econometrics*, 225(2):254–277.
- Roth, J., Sant’Anna, P. H. C., Bilinski, A., and Poe, J. (2023). What’s Trending in Difference-in-differences? A Synthesis of the Recent Econometrics Literature. *Journal of Econometrics*, 235(2):2218–2244.
- Sun, L. and Abraham, S. (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2):175–199.
- Wooldridge, J. M. (2025). Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators. *Empirical Economics*, forthcoming.

Table 1: Summary statistics for the simulated data set with 59,860 observations.

Variable	Mean	Std. Dev.	Minimum	Maximum
Outcome y	4.410	1.671	-2.425	13.751
If treated	0.0867	0.2814	0.0	1.0

Table 2: The expected value of the treatment effects by cohort and time.

Cohort	Time Periods								
	1	2	3	4	5	6	7	8	9
4	0.0	0.0	0.0	1.5	2.5	3.5	4.5	5.5	6.5
6	0.0	0.0	0.0	0.0	0.0	1.0	1.7	2.4	3.1
8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.9
Controls	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 3: The number of observations by cohort and time.

Cohort	Time Periods									Total
	1	2	3	4	5	6	7	8	9	
4	453	755	<i>1,359</i>	906	906	855	302	453	302	6,191
6	312	520	936	624	<i>624</i>	520	208	312	208	4,264
8	192	320	576	384	384	320	<i>128</i>	192	128	2,624
Controls	3,423	5,705	10,269	6,846	6,846	5,705	2,282	3,423	2,282	46,781
Total	4,380	7,300	13,140	8,760	8,760	7,300	2,920	4,380	2,920	59,860

Bold indicates observations in treated time periods. *Italics* indicates observations in the time period prior to the start of treatment.

Table 4: Estimated cohort-time treatment coefficients from DID models, and weights.

Cohort	Time	FLEX		Callaway Sant'Anna	
		Tx Effects	Weights	Tx Effects	Weights
4	4	1.540452	906	1.540452	906 +1359
4	5	2.452483	906	2.452483	906 +1359
4	6	3.534971	755	3.534971	755 +1359
4	7	4.581689	302	4.581689	302 +1359
4	8	5.500610	453	5.500610	453 +1359
4	9	6.517526	302	6.517526	302 +1359
6	6	0.995921	520	0.995921	520 +624
6	7	1.532444	208	1.532444	208 +624
6	8	2.276666	312	2.276666	312 +624
6	9	3.035713	208	3.035713	208 +624
8	8	0.662019	192	0.662019	192 +128
8	9	0.763577	128	0.763577	128 +128

The estimated cohort-time treatment effects for FLEX and Callaway and Sant'Anna are numerically identical for the models estimated and are close to the expected values (see Table 2). The weights used are different, resulting in different ATETs (see Table 5).

Table 5: Estimated overall ATETs from the DID models, based on estimated cohort-time treatment effects and weights in Table 5.

Weights	Estimated ATET	Std. Err.
FLEX	2.799203	(.1667929)
Callaway Sant'Anna	3.254948	(.1818035)

Table 6: Estimated event-time treatment effects.

Exposure	FLEX	Callaway Sant'Anna
-7	0.0114471	0.0114471
-6	0.0262384	0.0262384
-5	0.0712398	.0264314
-4	-0.0016993	-.0128961
-3	0.0133801	.0074171
-2	0.0445824	.043862
-1	0	0
0	1.261209	1.298016
1	2.124345	2.095241
2	3.167032	3.148816
3	3.951173	4.065743
4	5.500610	5.500610
5	6.517526	6.517526

The aggregated event-time treatment effects for FLEX and Callaway and Sant'Anna are similar, but not numerically identical up through exposure time 3. In exposure time periods 4 and 5 (shown in **bold**), the aggregation is only over the single cohort 4, so the aggregated event-time effects are the same because weights do not matter.

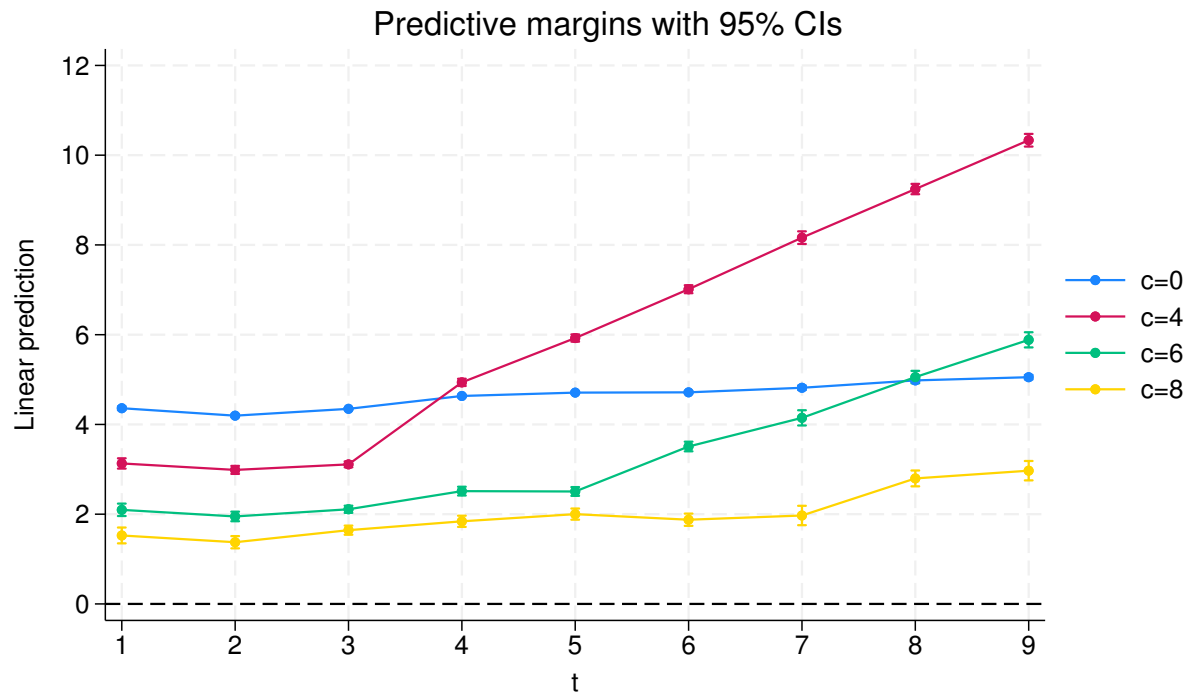


Figure 1: Predicted outcomes by cohort and time. Cohort 0 is the control.

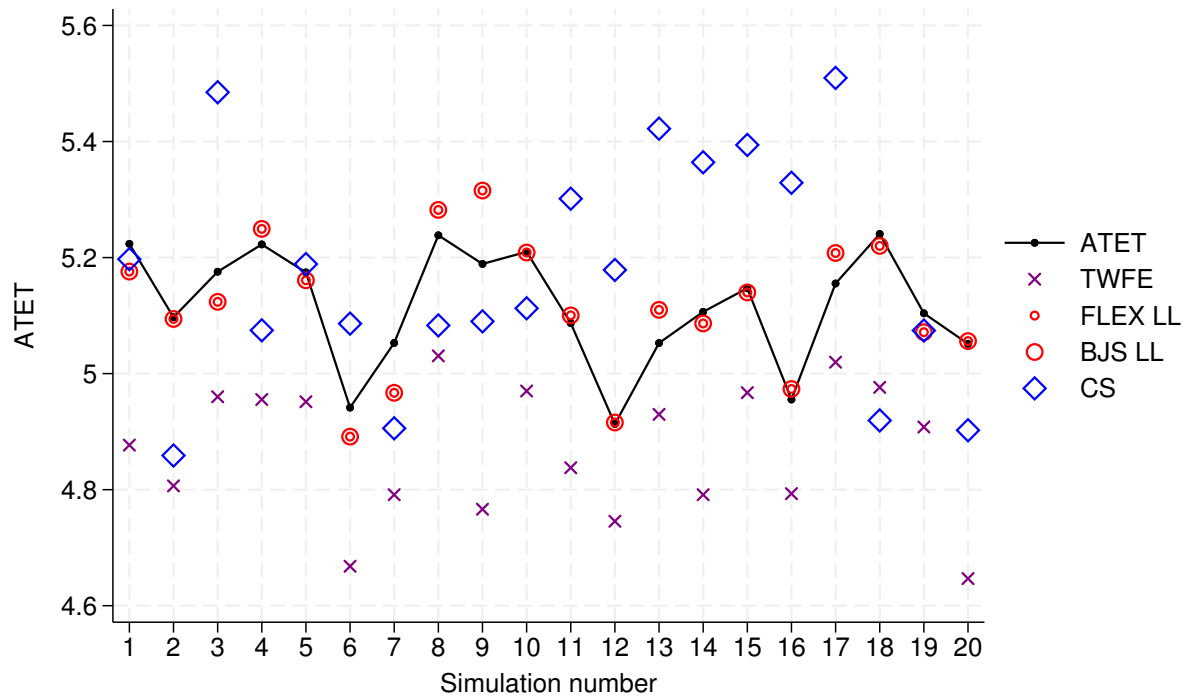


Figure 2: Simulated ATETs. All models except TWFE have identical estimates of cohort-time treatment effects.